General Relativity HW-1

PROBLEM 1. We have shown that an energy-momentum four-vector of a resting massive point particle is $p^\mu = (m, 0, 0, 0)$ This expression is written in units where the speed of light is $c = 1$. Show that if we keep c as a parameter one gets $p^\mu = (mc^2, 0, 0, 0)$.
Further show that after a boost to a system where our particle moves with a spatial velocity v one gets (if using $c = 1$) $p^\mu = (\gamma m, v\gamma m, 0, 0)$, for $\gamma = 1/\sqrt{1-v^2}$ and show that for a small vs standard expressions for a kinetic energy and momentum are restored.

SOLUTION:

By definition,$p^\mu = m\frac{dx^\mu}{d\tau}$(where τ is the proper time which satisfies $(d\tau)^2 = -c^2(dt)^2+(dx^i)^2$(when keeping parameter c)).When keeping parameter c,$x^\mu = (ct, x^i)$.Thus $p^\mu = m(c\frac{dt}{d\tau}, \frac{dx^i}{d\tau})$.Because we are considering a resting particle,$\frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = 0$.Thus $\frac{dt}{d\tau} = c$,and $p^\mu = m(c^2, 0, 0, 0) = (mc^2, 0, 0, 0)$.

For a standard boost along the first axis(which is generated by $K_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$,namely $\Lambda = e^{tK_1} = \begin{bmatrix} cosht & sinht & 0 & 0 \\ sinht & cosht & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$)

$\Lambda^\nu{}_\mu p^\mu = (cosh(t)m, sinh(t)m, 0, 0) = (\gamma m, v\gamma m, 0, 0)$,by letting $\gamma = cosh(t), v = tanh(t)$.

When v is small,by considering the Talor series of γ we have $\gamma = 1 + \frac{v^2}{2} + o(v^2)$.$p^\mu = (\gamma m, mv, 0, 0) + o(v^2)$,$E = \gamma m = m + \frac{mv^2}{2} + o(v^2), E_k = E|_v - E|_{v=0} = \frac{mv^2}{2} + o(v^2)$,which is reduced to the classical form.

PROBLEM 2. Consider a totally symmetric rank-3 tensor $T_{\mu\nu\rho}$. How many independent components does it have in D dimensions? In a simpler case, compute the number of independent components in 4 dimensions.

SOLUTION:

The independent components of symmetric tensor of rank-n,dimensions D is equivalent to the *stars and bars* problem,that counts the possible situations putting n identical balls into D different boxes.The assertion is proved by observing that each element of the tensor (which corresponds to an indices sequence) corresponds to a sequence of putting n different balls into D different boxes (choosing a dimension for each index).But due to the symmetry,the order of the indices doesn't matter as long as you put the same number of indices into each dimension,which is equivalently saying that the balls are all identical.

It is well-known that the solution to *stars and bars* problem is $\binom{n+D-1}{n}$.Thus a rank-3 tensor in D dimensions has $\binom{D+2}{3}= \frac{(D+2)(D+1)D}{6}$.When $D = 4$,there are 20 independent components.

PROBLEM 3. Consider a totally anti-symmetric rank-3 tensor $T_{\mu\nu\rho}$. How many independent components does it have in D dimensions? In a simpler case, compute the number of independent components in 4 dimensions.
Prove that there is no nontrivial totally anti-symmetric tensor of rank $D+1$ in a D dimensional space-time.

SOLUTION:

For any element of T that has at least 2 same indices ($T_{1,1,2}$ for instance),then permute the 2 same indices and we found that it is equal to its opposite,which is 0.Thus all non-trivial component in T must have different indices.The total number is thus $\binom{D}{3}= \frac{D(D-1)(D-2)}{6}$.For $D = 4$,there are in total 4 independent components.

For any totally anti-symmetric tensor of rank $D + 1$ in D dimensional space,by pigeonhole principle,for each component of T (and correspondingly each sequence of indices),there exists at least 2 indices that are the same (in the same dimension).Then exchange the 2 indices gives you that it is 0.Thus T is trivial since all of its components are 0.